

Configuration Commands

Lion_R:

```
enable
configure terminal
interface GigabitEthernet0/0
ip address 10.0.12.129 255.255.255.240
no shutdown
interface Serial0/0/0
ip address 10.0.12.145 255.255.255.252
no shutdown
ip route 10.0.12.0 255.255.255.128 10.0.12.146
ip route 10.0.8.0 255.255.252.0 10.0.12.146
ip route 10.0.0.0 255.255.248.0 10.0.12.146
end
write memory
```

```
enable
configure terminal
ip route 10.0.12.0 255.255.255.128 10.0.12.146
ip route 10.0.8.0 255.255.252.0 10.0.12.146
ip route 10.0.0.0 255.255.248.0 10.0.12.146
end
```

write memory

Elephant_R:

```
enable
configure terminal
interface GigabitEthernet0/0
ip address 10.0.12.1 255.255.255.128
ip helper-address 10.0.12.134
no shutdown
interface Serial0/0/0
ip address 10.0.12.146 255.255.255.252
no shutdown
interface Serial0/0/1
ip address 10.0.12.149 255.255.255.252
no shutdown
interface Serial0/1/0
```

```
ip address 10.0.12.153 255.255.255.252
no shutdown
ip route 10.0.12.128 255.255.255.240 10.0.12.145
router rip
version 2
no auto-summary
network 10.0.12.0
redistribute static
end
```

```
write memory
```

```
enable
```

```
configure terminal
```

```
interface GigabitEthernet0/0
```

```
ip helper-address 10.0.12.134
```

```
end
```

```
write memory
```

Cat_R:

```
enable
configure terminal
interface GigabitEthernet0/0
ip address 10.0.8.1 255.255.252.0
ip helper-address 10.0.12.134
no shutdown
interface Serial0/0/0
ip address 10.0.12.150 255.255.255.252
no shutdown
router rip
version 2
no auto-summary
network 10.0.8.0
network 10.0.12.0
end
```

```
write memory
```

enable

configure terminal

interface GigabitEthernet0/0

ip helper-address 10.0.12.134

end

write memory

Dog_R:

enable

configure terminal

interface GigabitEthernet0/0

ip address 10.0.0.1 255.255.248.0

ip helper-address 10.0.12.134

no shutdown

interface Serial0/0/0

ip address 10.0.12.154 255.255.255.252

no shutdown

router rip

version 2

no auto-summary

network 10.0.0.0

network 10.0.12.0

end

write memory

enable

configure terminal

interface GigabitEthernet0/0

ip helper-address 10.0.12.134

end

write memory

VLSM Tree

Starting block: 10.0.0.0/20 (4096 addresses).

- Subnet 1 (Dog LAN): 10.0.0.0/21 (2048 addresses, supports 2046 hosts for pop 1200 + overhead).
- Subnet 2 (Cat LAN): 10.0.8.0/22 (1024 addresses, supports 1022 hosts for pop 800 + overhead).
- Subnet 3 (Elephant LAN): 10.0.12.0/25 (128 addresses, supports 126 hosts for pop 100 + overhead).
- Subnet 4 (Lion LAN): 10.0.12.128/28 (16 addresses, supports 14 hosts for pop 3 + chief + 4 servers + 2 reps).
- Subnet 5 (Link Lion-Ele): 10.0.12.144/30 (4 addresses, 2 hosts).
- Subnet 6 (Link Ele-Cat): 10.0.12.148/30 (4 addresses, 2 hosts).
- Subnet 7 (Link Ele-Dog): 10.0.12.152/30 (4 addresses, 2 hosts).

IP Address Table

Device	Interface/Port	IP Address	Subnet Mask	Default Gateway
R_Lion	GigabitEthernet0/0	10.0.12.129	255.255.255.240	N/A
R_Lion	Serial0/0/0	10.0.12.145	255.255.255.252	N/A
R_Elephant	GigabitEthernet0/0	10.0.12.1	255.255.255.128	N/A
R_Elephant	Serial0/0/0	10.0.12.146	255.255.255.252	N/A
R_Elephant	Serial0/0/1	10.0.12.149	255.255.255.252	N/A
R_Elephant	Serial0/1/0	10.0.12.153	255.255.255.252	N/A
R_Cat	GigabitEthernet0/0	10.0.8.1	255.255.252.0	N/A
R_Cat	Serial0/0/0	10.0.12.150	255.255.255.252	N/A
R_Dog	GigabitEthernet0/0	10.0.0.1	255.255.248.0	N/A
R_Dog	Serial0/0/0	10.0.12.154	255.255.255.252	N/A
Chief_Lion_PC	Ethernet	10.0.12.130	255.255.255.240	10.0.12.129
Lion_PC1	Ethernet	10.0.12.135	255.255.255.240	10.0.12.129
Lion_PC2	Ethernet	10.0.12.136	255.255.255.240	10.0.12.129
Web_Server	Ethernet	10.0.12.131	255.255.255.240	10.0.12.129

DNS_Server	Ethernet	10.0.12.132	255.255.255.240	10.0.12.129
Email_Server	Ethernet	10.0.12.133	255.255.255.240	10.0.12.129
DHCP_Server	Ethernet	10.0.12.134	255.255.255.240	10.0.12.129
Chief_Elephant_PC	Ethernet	10.0.12.2	255.255.255.128	10.0.12.1
Elephant_PC1	Ethernet	DHCP Assigned	DHCP Assigned	DHCP Assigned
Elephant_PC2	Ethernet	DHCP Assigned	DHCP Assigned	DHCP Assigned
Chief_Cat_PC	Ethernet	10.0.8.2	255.255.252.0	10.0.8.1
Cat_PC1	Ethernet	DHCP Assigned	DHCP Assigned	DHCP Assigned
Cat_PC2	Ethernet	DHCP Assigned	DHCP Assigned	DHCP Assigned
Chief_Dog_PC	Ethernet	10.0.0.2	255.255.248.0	10.0.0.1
Dog_PC1	Ethernet	DHCP Assigned	DHCP Assigned	DHCP Assigned
Dog_PC2	Ethernet	DHCP Assigned	DHCP Assigned	DHCP Assigned